Economic Fibrillation — Crisis-Window Synchronization Analysis

Measuring the Kuramoto order parameter of 26 economies (World Bank GDP, current US$) around the four truly global crises of 1970–2022

1 Executive summary

We measured business-cycle synchronization *r(t)* for every year from 1970 to 2022 and examined its behaviour around the four crises in which essentially the whole world contracted together (the 1973–75 and 1980–82 oil shocks, the 2007–09 Global Financial Crisis, and 2020 COVID-19). Three findings stand out. **First**, synchronization during global-crisis years is higher on average (r = 0.75) than in normal years (r = 0.67), and a relative test places this elevation beyond chance (p = 0.038). **Second**, regional crises show the opposite: they coincide with the *lowest* synchronization in the entire record (r = 0.45 on average), confirming that r cleanly separates truly global shocks from regional ones. **Third**, the highest synchronization of all (0.91, in 2019) occurred in a year with no global crisis — so high synchronization is a necessary precondition for global contagion (a 'loaded gun'), not a guarantee of it.

2 What we did (method and formulas)

Each economy's GDP rises over time, so it is not itself an oscillator. We therefore separate the rising trend from the business-cycle wobble, and measure where each economy sits in its cycle. The pipeline, applied identically to all economies, is:

(i) **log-linearise**: $g(t) = \ln G(t)$, turning exponential growth into a near-straight trend.

(ii) **band-pass detrend**: keep only fluctuations of business-cycle length (≈ 2–9 years) with a 2nd-order Butterworth filter → the cyclical part x(t).

(iii) **Hilbert transform**: form the analytic signal $z(t) = x(t) + i\cdot H[x](t) = A(t)\cdot e^{i\theta(t)}$; its angle θ(t) is each economy's cycle phase (the 'clock-hand' position).

(iv) **Kuramoto order parameter**: $r(t)\cdot e^{i\psi(t)} = (1/N)\,\Sigma_j\, e^{i\theta_j(t)}$. r ranges from 0 (economies scattered) to 1 (perfectly locked). r(t) is the single number this report tracks.

This is measurement, not simulation, and it uses no machine learning: r is read directly off the data. Parameters: N = 26 economies; filter band ≈ 2–9 yr, order 2; annual sampling. The filter was run on the longest available record (1960–2023 for 25 of 26 economies) so that the 1973–75 window sits well clear of filter edge effects.

3 Overview: synchronization across five decades

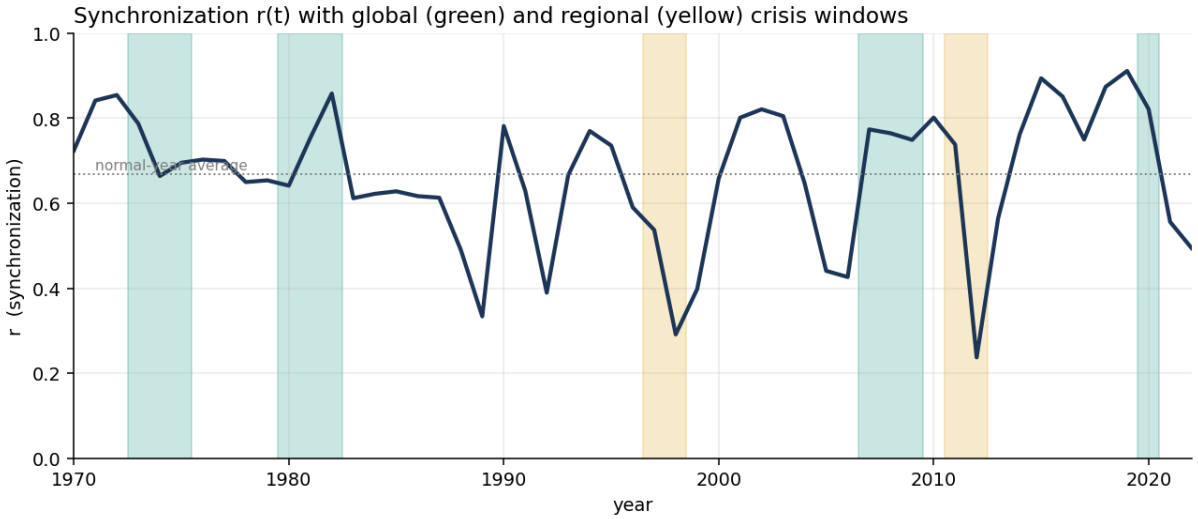


Figure 1. Synchronization r(t), 1970–2022. Green bands mark the four global crises; yellow bands mark two regional crises (Asian 1997–98, Euro 2011–12). Global crises sit at or above the normal-year average; regional crises sit at the troughs.

4 Zoom: each global crisis



Figure 2. Synchronization in a ±4-year window around each global crisis (crisis years shaded).

Oil shock II / Volcker (1980–82) is the textbook case: r *rises* straight into the crisis, from 0.64 to 0.75 to 0.86 — the last of these in the 92th percentile of all years. **The Global Financial Crisis (2007–09)** shows a sustained high plateau (r ≈ 0.77–0.75), having climbed from the mid-2000s lows (r ≈ 0.44 in 2005–06). **COVID-19 (2020)** is high during the shock (r = 0.82, 85th percentile) and then *shatters* afterwards — falling to 0.56 in 2021 and 0.49 in 2022, the clearest post-crisis desynchronisation in the record. **Oil shock I (1973–75)** is the least clean: synchronization entered high (0.79) but eased during the shock; note this is also the earliest window and the most sensitive to data limitations.

5 Composite shape around crises

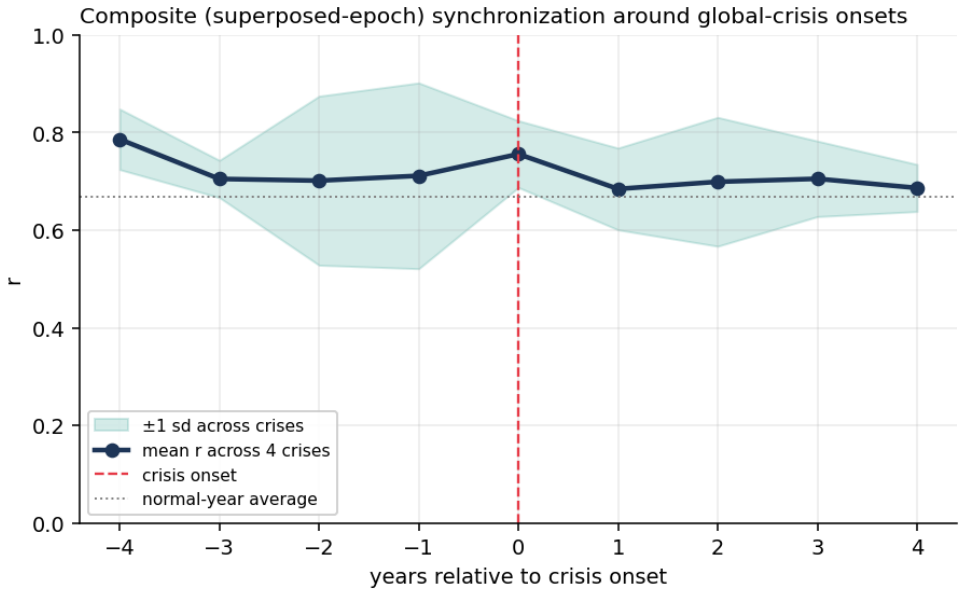


Figure 3. Superposed-epoch (composite) analysis: r averaged across all four crises, aligned at onset (year 0). On average, synchronization is elevated above the normal-year baseline through the onset window, consistent with the 'fragility builds into the crisis' picture, though the four crises differ individually.

6 Relative test: is the elevation real?

Because synchronization is high and persistent throughout (a 'world business cycle'), the right question is not 'is r above zero-coupling chance' (it almost always is) but 'is r during the actual global crises higher than you would get by placing the same crisis windows at random times'. We test this by circularly shifting the crisis-window pattern to 20,000 random positions and comparing.

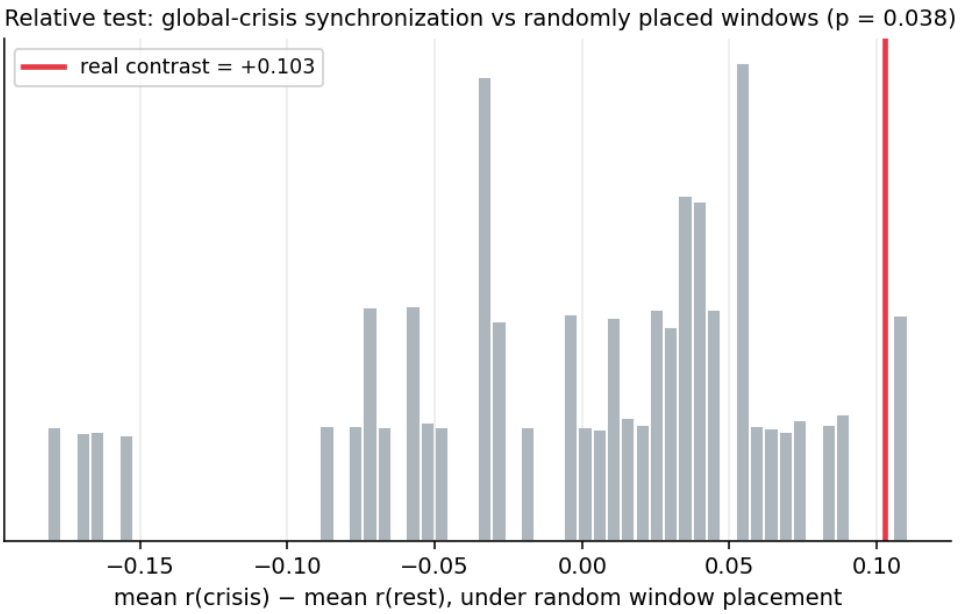


Figure 4. The real global-crisis synchronization contrast (+0.103) sits in the upper tail of the random-placement distribution: p = 0.038. The elevation is statistically significant at the 5% level.

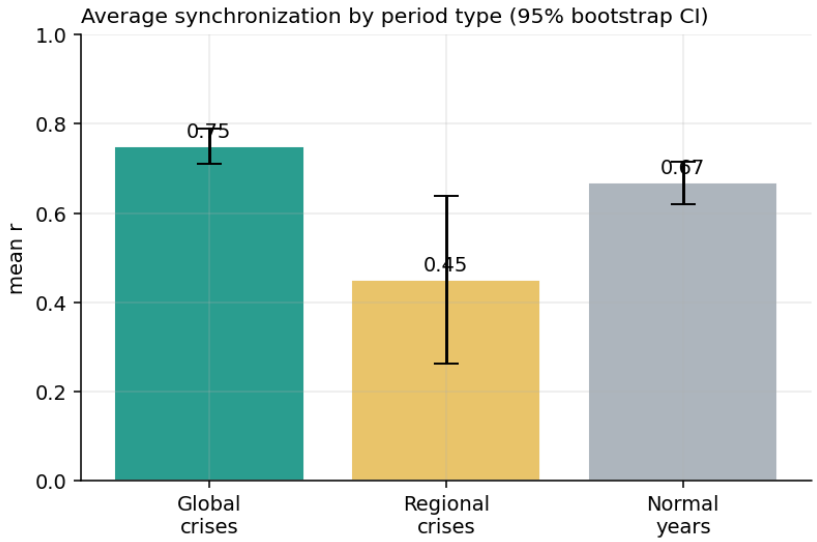


Figure 5. Mean synchronization by period type (95% bootstrap intervals). Global crises (0.75) > normal years (0.67) >> regional crises (0.45).

7 The regional-crisis contrast (a validation)

Regional crises are where the method proves it is measuring something real. When one region collapses while others grow, the world is *desynchronised* — and the data agrees emphatically:

Regional crisis year	r	percentile among all years
Asian crisis 1997	0.54	17th
Asian crisis 1998	0.29	2th (near lowest)
Euro debt crisis 2011	0.74	58th
Euro debt crisis 2012	0.24	0th (lowest in record)

8 Key observations

- Global-crisis years are, on average, the most synchronized (r = 0.75), significantly above normal years (r = 0.67; p = 0.038).
- Regional crises coincide with the lowest synchronization in the record (Asian 1998 and Euro 2012 near the very bottom), showing r distinguishes global from regional shocks.
- Synchronization visibly builds into the 1980–82 and 2007–09 crises, and shatters after 2020 — the rise-then-rupture pattern at the heart of the ‘economic fibrillation’ idea.
- High synchronization is necessary but not sufficient: the peak year (2019, r = 0.91) had no global crisis. Fragility loads the gun; a shock still pulls the trigger.

9 Limitations (stated plainly)

- GDP is in current US dollars, so year-to-year changes carry some inflation and exchange-rate movement; the band-pass removes the slow part of this but not short-lived currency swings. A constant-price (real GDP) robustness run is a natural next step.
- Only four global crises exist in the record, and the data is annual, so statistical power is limited; p = 0.038 is significant but not overwhelming, and individual crises differ.

- The 2020 window sits near the end of the data, where the filter is less reliable; the very large COVID contraction still dominates.
- Synchronization is autocorrelated; the relative test is designed to respect this by shifting whole windows rather than single years.

10 Conclusion and next step

The measurement supports the core of the economic-fibrillation thesis: the world's economies lock together during truly global crises, significantly more than in ordinary years and far more than during regional crises, and this locking builds into and ruptures around the largest shocks. It also disciplines the claim: synchronization is a state of fragility — a necessary precondition for global contagion — rather than a trigger. The natural next step is the coupling-collapse simulation (the Kuramoto differential equation with a falling coupling K), which would show the synchronized cluster shattering as coupling breaks — the mechanism this measurement can only observe from the outside.